

Research and Applications

Physician understanding, explainability, and trust in a hypothetical machine learning risk calculator

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ABSTRACT

Objective: Implementation of machine learning (ML) may be limited by patients' right to "meaningful information about the logic involved" when ML influences healthcare decisions. Given the complexity of healthcare decisions, it is likely that ML outputs will need to be understood and trusted by physicians, and then explained to patients. We therefore investigated the association between physician understanding of ML outputs, their ability to explain these to patients, and their willingness to trust the ML outputs, using various ML explainability methods.

Materials and Methods: We designed a survey for physicians with a diagnostic dilemma that could be resolved by an ML risk calculator. Physicians were asked to rate their understanding, explainability, and trust in response to 3 different ML outputs. One ML output had no explanation of its logic (the control) and 2 ML outputs used different model-agnostic explainability methods. The relationships among understanding, explainability, and trust were assessed using Cochran-Mantel-Haenszel tests of association.

Results: The survey was sent to 1315 physicians, and 170 (13%) provided completed surveys. There were significant associations between physician understanding and explainability ($P < .001$), between physician understanding and trust ($P < .001$), and between explainability and trust ($P < .001$). ML outputs that used model-agnostic explainability methods were preferred by 88% of physicians when compared with the control condition; however, no particular ML explainability method had a greater influence on intended physician behavior.

Conclusions: Physician understanding, explainability, and trust in ML risk calculators are related. Physicians preferred ML outputs accompanied by model-agnostic explanations but the explainability method did not alter intended physician behavior.

Key words: artificial intelligence, explainability, interpretability, decision support, medicine

INTRODUCTION

BACKGROUND AND SIGNIFICANCE

In clinical practice, traditional risk calculators, such as the Wells' Criteria and PERC Rule scores for pulmonary embolism, are commonly

used in clinical practice to support decision making in cases of diagnostic uncertainty.^{1–3} Machine learning (ML) may improve the accuracy of risk assessment⁴; however, implementation of ML raises complex clinical, ethical, and legal questions because of the lack of un-

derstanding about how ML models generate outputs—commonly referred to as the black box problem.^{5–7} This has become a particular issue since the European Union Global Data Protection Regulation has been introduced, which requires that patients have a right to “meaningful information about the logic involved” when ML is used.⁸ Therefore, unless the black box problem can be remedied, the patient right to explanation cannot be fulfilled and the healthcare industry cannot fully benefit from these powerful technologies.

Fortunately, ML explainability methods have been developed to address the black box problem.^{9–13} For example, model-agnostic explainability methods can derive post hoc explanations from black box models.¹⁴ Broadly, this involves training an interpretable model based on the predictions of a black box model and perturbing its inputs to see how the black box model reacts.^{15–17} Examples of these methods include permutation variable importance (VI), individual conditional expectation (ICE) plots, local interpretable model-agnostic explanations (LIME), and Shapley values (SVs).^{17–20} While these methods have been explored in a number of contexts, there is currently limited evidence as to whether they adequately explain ML-derived risk estimates in a clinical context.²¹

Importantly, clinical decisions represent a complex synthesis of basic sciences, clinical evidence, and patient preferences.^{22,23} If these highly individualized decisions were to be automated with ML, physicians would likely need to explain the decision-making process to patients, at least in the near term.^{22,24,25} Indeed, many published examples of ML applications in healthcare still require physician interpretation.^{26–30} Therefore, in order to satisfy patients’ rights to a meaningful explanation, ML risk calculators will first need to “explain” their output in such a way that physicians can understand. It is not yet known if currently available model-agnostic explanations can provide physicians with an adequate understanding of ML outputs, nor if physician understanding of ML outputs leads to the ability to explain them to patients. Finally, few studies have explored whether improving physician understanding of ML outputs influences physician behavior.

OBJECTIVES

In the context of a clinical ML risk calculator, we aimed to investigate the association among (1) physician understanding of an ML output (termed physician understanding); (2) physician ability to explain an ML output to patients (termed explainability), and (3) and physician intended behavior (termed trust). We also aimed to explore whether model-agnostic explainability methods influenced the intended behavior of physicians, and whether a particular model-agnostic explainability method was preferred by physicians.

MATERIALS AND METHODS

Hypothetical ML risk calculator for pulmonary embolism

To assess the relationships among physician understanding, explainability, and trust in ML risk calculators, we designed a hypothetical ML risk calculator based on the Wells’ Criteria and PERC Rule to use in a survey-based clinical scenario for physicians.^{1,2} Based on the risk estimated by the ML risk calculator, a clinical recommendation of either (1) “reassurance and discharge recommended” or (2) “computed tomography pulmonary angiogram recommended” was made if the patient was low risk of pulmonary embolism or not low risk of pulmonary embolism, respectively. We combined a risk esti-

mate with a clinical recommendation as we believed that it would allow us to assess intended physician behavior and, therefore, trust in the ML output. In addition, the combination of a risk estimate and a clinical recommendation is commonly used with non-ML risk calculators, including online versions of the Wells’ Criteria and PERC Rule.^{31,32}

Survey instrument

Based on the described hypothetical ML risk calculator, we designed an online survey ([Supplementary Appendix](#)) that provided participants with the following hypothetical clinical scenario:

You are a GP [general practitioner] who has reviewed a 50-year-old woman presenting with shortness of breath. After a history, examination, laboratory tests, ECG [electrocardiogram], and a chest x-ray, you are comfortable you have excluded the most concerning diagnoses. However, you are still considering pulmonary embolism. The practice has installed a piece of software that uses artificial intelligence to assist with ruling out pulmonary embolism. It can stratify patients as either; (1) low risk of pulmonary embolism: reassurance and discharge recommended; or (2) not low risk of pulmonary embolism: computed tomography pulmonary angiogram recommended. The software automatically analyses the electronic record, including your documented history, examination, and laboratory tests, and provides its recommendation.

The rationale for this clinical scenario was that pulmonary embolism is a disease encountered in community, medical, and surgical settings and is potentially life-threatening, so decision making around diagnosis and management is critical.³³ We provided limited clinical information in the scenario to prevent participants from agreeing or disagreeing with the ML output for reasons other than the explainability method we used. Each participant was shown 3 consecutive ML outputs, which included an ML risk estimate combined with a clinical recommendation. The 3 consecutive ML outputs were identical except for the method used to visualize and explain the “logic” of the risk estimate. After each ML output, the participant was asked the following questions with select-choice answers:

1. To what degree does the software’s decision make sense to you? [Not at all/Very little/Somewhat/To a great extent]
2. Would you be able to explain the software’s decision to the patient? [Yes/No]
3. Would you follow the software’s recommendation? [Yes/No]

The first ML output for all participants was as follows:

Your patient has a low risk (<1% chance) of pulmonary embolism. They should be reassured and followed up in the community as you deem appropriate. This recommendation is based on a cohort of 10 000 patients who were investigated for pulmonary embolism, of whom 1000 had a similar risk profile. The software has been externally validated.

This ML output was regarded as the control condition because only the estimated risk of pulmonary embolism (ie, <1% chance) and the population that the ML model was derived from were shown. No explanation of the logic behind the ML model’s risk calculation was shown. In our scenario, the accuracy of the ML model and size of the derivation cohort were exaggerated in order to ensure that physicians’ potential lack of trust was not simply due to poor model performance. For the second and third ML outputs, the control output was shown to the participant along with a model-agnostic explanation. This model-agnostic explanation included (1) a graphical visualization and (2) a brief textual explanation of

what the visualization was demonstrating. The model-agnostic explanations were hypothetical. They were developed by W.K.D. and N.B., who are practicing physicians, and Q.T., who is a data scientist, and were tested on a small number of practicing physicians in order to ensure they were clinically valid.

For the second ML output, participants were randomized to 1 of 2 global model-agnostic explanations, either VI (Figure 1A) or an ICE plot (Figure 1B). For the third ML output, participants were randomized to 1 of 2 local model-agnostic explanations, either LIME (Figure 1C) or SVs (Figure 1D). Broadly, global explanations demonstrate feature importance across a population, while local explanations explain individual predictions.³⁴ Finally, participants were asked to select which ML output they preferred and why. Demographics and medical subspecialty were also assessed.

Participants

The relevant institutional review boards approved this study. We surveyed all physicians employed in Northland District Health Board (approximately 104 Resident Medical Officers and 150 Senior Medical Officers), Waitematā District Health Board (approximately 415 Resident Medical Officers and 502 Senior Medical Officers), and Te Tai Tokerau and Manaia Primary Health Organisations (approximately 151 general practitioners combined) in New Zealand between March and May of 2019. These physicians serve a population of approximately 800 000 people. The survey was emailed to all physicians via a SurveyGizmo link.³⁵

Descriptive statistics and analysis of comments

Statistical analysis was performed using data analysis (Pandas and StatsModels) and visualizations packages (Seaborn and WordCloud).

in the Python ecosystem, in addition to stats and sampleSizeCMH in the R ecosystem. The associations between physician understanding, explainability, and trust were assessed using Cochran-Mantel-Haenszel tests. Because repeated measures were collected from each participant (ie, 1 for each of the 3 ML output types), the response data were stratified according to the ML output type (ie, control condition, global explanation, and local explanation). Thereby, $2 \times 4 \times 3$ contingency tables were generated for physician understanding vs explainability and for physician understanding vs trust, and $2 \times 2 \times 3$ contingency tables were generated for explainability vs trust. For $2 \times J \times K$ ($J > 2$) tables, the generalized Cochran-Mantel-Haenszel test was applied.³⁶ To assess the reliability of each test, the statistical power of each test were calculated with a conventional significance level of .05.

Binomial exact tests were used to assess whether physicians preferred specific explainability methods. To assess the influence of explainability method on intended physician behavior (trust), McNemar's test was applied to paired responses from the same participants under 2 different scenarios (eg, control vs global explanation). If the number of changes in responses was < 25 , a binomial distribution was used to obtain an exact P value, otherwise a normal chi-square distribution was used to achieve an approximated P value.

A word cloud was used to explore the reasons why participants preferred certain ML outputs. Natural language processing package was used to process the responses. The plain text was tokenized, decapitalized, and stemmed. After removing all punctuation, digits, stop words, and short words with < 3 letters, the frequency of each token and bigram was counted and used to set the font size in the word cloud proportionally. We kept only tokens or bigrams that appear at least 5 times in the responses collected.

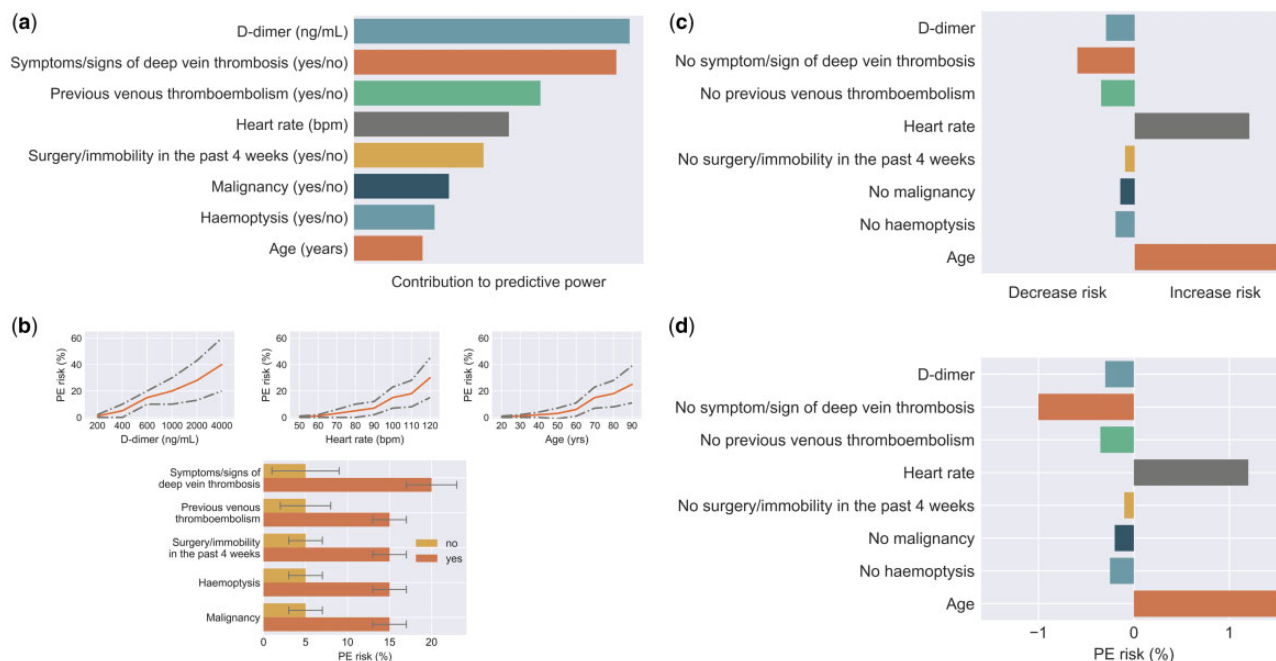


Figure 1. Graphical visualizations shown to participants with the following complementary explanation. (A) For variable importance (VI), the visualization shows the relative importance of each clinical factor used by the model. This is general information about the software's logic. (B) For individual conditional expectation plots (ICE), the visualizations show the average and standard deviation of the software's predictions for different values of the most influential clinical factors used by the model. This is general information about the software's logic. (C) For local interpretable model-agnostic explanations (LIME), the visualization shows the positive or negative relative impact of the most influential clinical factors used by the software to estimate your patient's risk of pulmonary embolism (PE). This profile is specific to your patient. (D) For Shapely values (SVs), the visualization shows the positive or negative contribution of each clinical factor to the risk estimated by the software. The sum of the bars is equal to your patient's PE risk. This profile is specific to your patient.

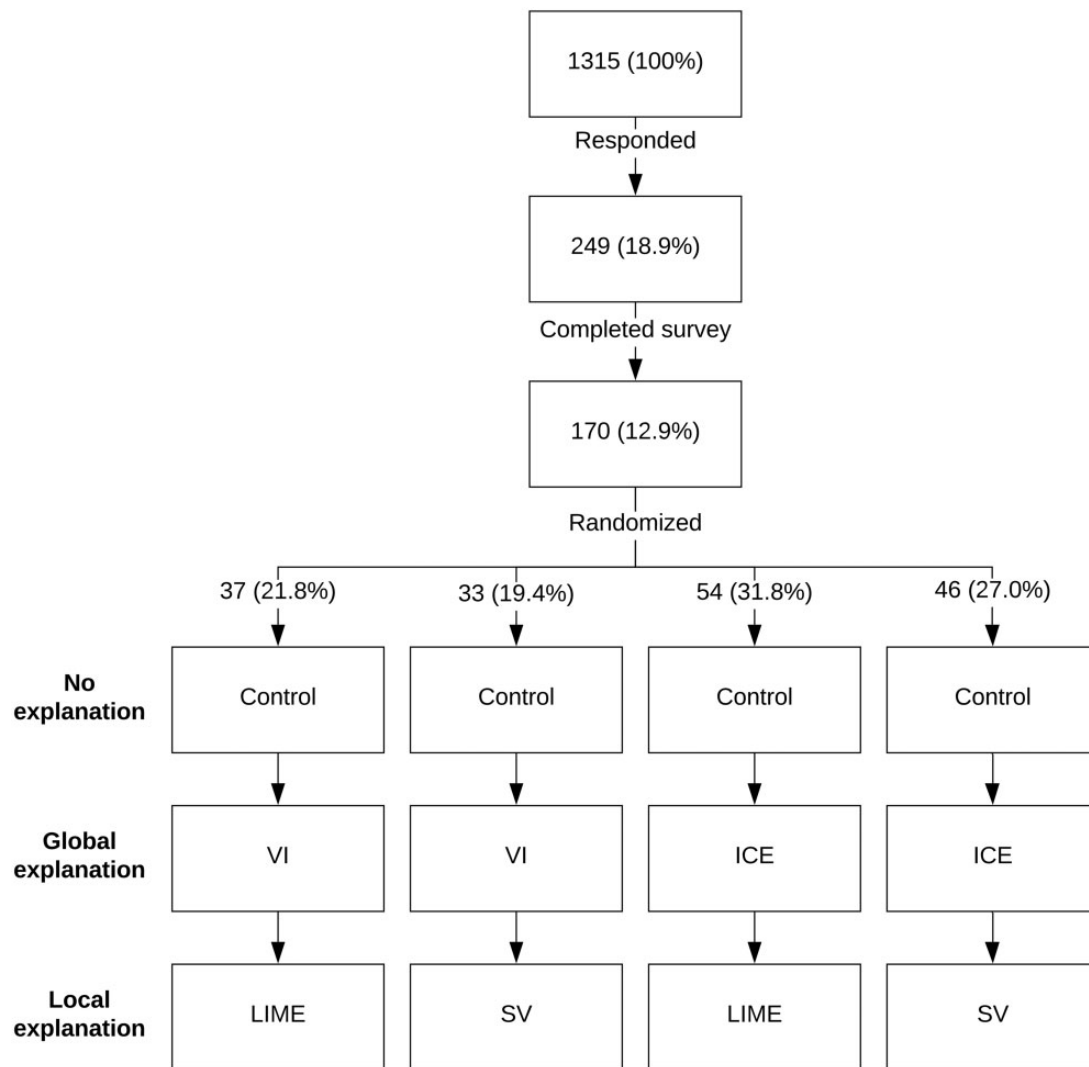


Figure 2. Flow chart of survey response and randomization. ICE: individual conditional expectation plots; LIME: local interpretable model-agnostic explanations; SV: Shapely values; VI: variable importance

RESULTS

Participants

The survey was emailed to 1315 participants (Figure 2). There were 249 (18.9%) responses, among which 79 (31.7%) were incomplete and 170 (68.3%) were complete. A total of 100 were male participants (58.8%). The largest age group was the 25-34 years of age range (32.4%) followed by the 45-54 years of age range (27.6%). The most common medical subspecialties that responded were “Medicine” (20%) followed by “Other” (17.6%) and “Anaesthetics” and “Psychiatry” (both 11.8%) (Table 1). Of the 170 physicians that completed the survey, all (100%) were provided with 3 consecutive ML outputs, of which the first was always the control output (no model-agnostic explanation), the second was always a global model-agnostic explanation (VI or ICE), and the third was always a local model-agnostic explanation (LIME or SVs). A total of 37 (21.8%) were randomized to VI and LIME; 33 (19.4%) were randomized to VI and SVs; 54 (31.8%) were randomized to ICE and LIME; and 46 (27.0%) were randomized to ICE and SVs. A total of 510 responses (3 ML outputs per physician who com-

pleted the survey) were used to assess the relationships among the concepts of physician understanding, explainability, and trust.

Physician understanding and explainability

Physicians who reported higher levels of understanding in response to the question “To what degree does the software’s decision make sense to you?” were more likely to answer “Yes” to the question “Would you be able to explain the software’s decision to the patient?” (Figure 3). The relationship between physician understanding and explainability was statistically significant, controlling for all ML output types (Cochran-Mantel-Haenszel $\chi^2_3 = 156.3$, $P < .001$, power = 0.99).

Physician understanding and trust

Physicians who reported higher levels of understanding in response to the question “To what degree does the software’s decision make sense to you?” were more likely to answer “Yes” to the question “Would you follow the software recommendation?” (Figure 4). The relationship between physician understanding and trust was

Table 1. Demographic characteristics of the participants

Characteristics	n (%)
Age	
18-24 y	3 (1.8)
25-34 y	55 (32.4)
35-44 y	34 (20.0)
45-54 y	47 (27.6)
55-64 y	22 (12.9)
65-74 y	5 (2.9)
75 y or older	3 (1.8)
Missing	1 (0.6)
Sex/gender	
Male	100 (58.8)
Female	64 (37.6)
Nonbinary	1 (0.6)
Missing	5 (2.9)
Subspecialty	
Medicine	34 (20)
Other	30 (17.6)
Anesthetics	20 (11.8)
Psychiatry	20 (11.8)
Surgery	16 (9.4)
Emergency medicine	15 (8.8)
Pediatrics	11 (6.5)
Radiology	9 (5.3)
General practice	6 (3.5)
Obstetrics and gynecology	5 (2.9)
Pathology	3 (1.8)
Ophthalmology	1 (0.6)

statistically significant, controlling for all ML output types (Cochran-Mantel-Haenszel $\chi^2_3 = 128.7$, $P < .001$, power = 0.99).

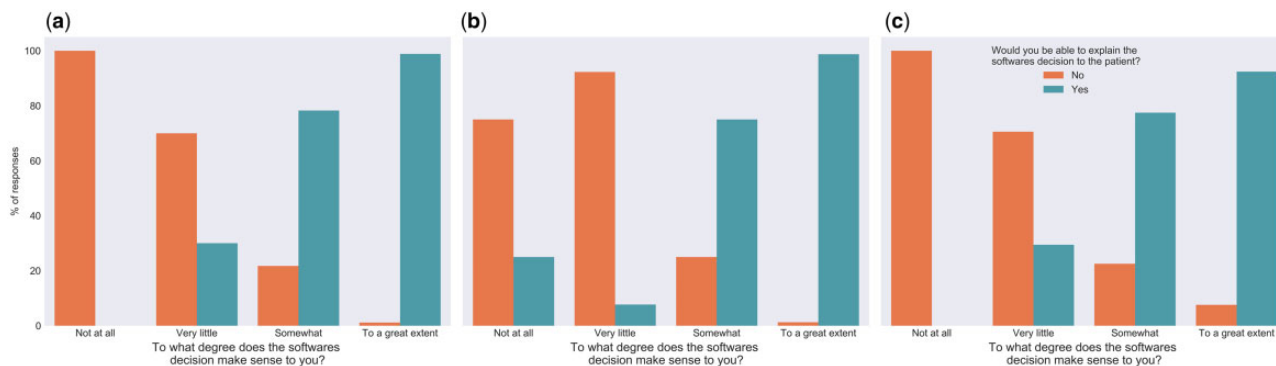
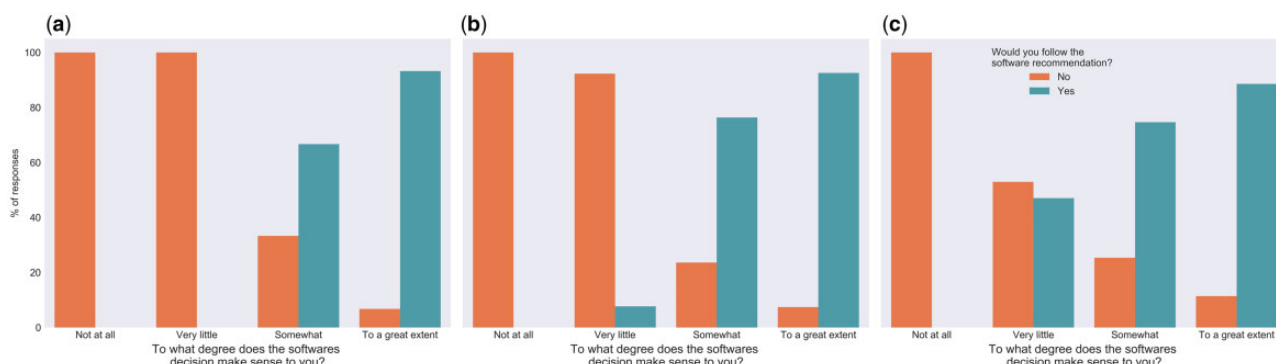
Explainability and trust

Physicians who reported “Yes” to the question “Would you be able to explain the software’s decision to the patient” were more likely to respond “Yes” to the question “Would you follow the software recommendation?” (Figure 5). The relationship between explainability and trust was statistically significant, controlling for all ML output types (Cochran-Mantel-Haenszel $\chi^2_1 = 61.2$, $P < .001$, power = 0.99). No particular ML explainability method had a greater influence on intended physician behavior (Table 2).

Preferred explainability method

There were 156 (91.8%) physicians that rated their preferred explainability method. Of these, 87.8% of physicians preferred an ML output which contained a model-agnostic explanation, compared with 12.2% who preferred the control output (no model-agnostic explanation). Binomial exact testing showed that the preference for any model explainability method was significantly higher than two-thirds ($P < .001$). Local explanations (LIME [32.1%] and SVs [29.9%]) were preferred over global explanations (VI [18.2%] and ICE [19.7%]), and binomial exact testing showed that the preference for local explanations (62.0%) was significantly higher than 0.5 among all those preferred model-agnostic explainability method ($P = .003$).

Even when no model-agnostic explanation was provided (ie, before being shown a model-agnostic explanation), physicians

**Figure 3.** Relationship between physician understanding and explainability. (A) Control condition. (B) Global explanation. (C) Local explanation.**Figure 4.** Relationship between physician understanding and trust. (A) Control condition. (B) Global explanation. (C) Local explanation.

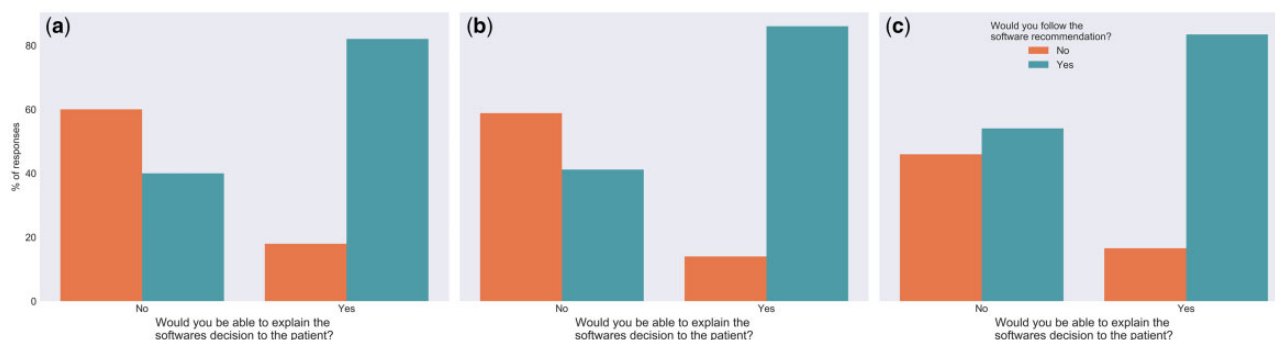


Figure 5. Relationship between explainability and trust. (A) Control condition. (B) Global explanation. (C) Local explanation.

Table 2. Influence of explainability methods on intended physician behavior

Concept	Compared scenarios	Observations	P value
Trust	Control Global explanation	170	.851
	Control Local explanation	170	.856
	Control VI	70	.774
	Control ICE	100	1.0
	Control LIME	91	.454
	Control SVs	79	.180
	VI LIME	37	.109
	VI SVs	33	.625
	ICE LIME	54	.754
	ICE SVs	46	.070

ICE: individual conditional expectation; LIME: local interpretable model-agnostic explanations; SV: Shapley value; VI: variable importance.

commonly reported that ML outputs were understandable (93% rating “somewhat” or “to a great extent”), explainable (85%), and trustworthy (76%). Figure 6 shows a word cloud generated from the responses to the question “Why do you prefer this software recommendation?” We noticed trends toward (1) the simplicity of the visualization (easy, simple, easy to understand), (2) the specificity of the explanation (specific, specific patient), and (3) the confirmation of clinical knowledge (explain, understand, information).

DISCUSSION

In the context of an ML risk calculator, this study found that physicians’ perceived understanding of ML outputs (physician understanding), their ability to explain ML outputs to a patient (explainability), and their intended behavior (trust) are significantly related. While physicians did prefer to have an explanation of the ML model’s logic (88% of physicians), no particular ML explainability method had a greater influence on intended physician behavior. These findings suggest that ML explainability is important to physicians, but that factors other than explainability, such as ML accuracy, might be more important to physicians. This hypothesis needs to be tested in future studies.

Physician understanding and intended physician behavior (trust)

Several other studies have explored the link between physician understanding and trust in ML. A survey of 191 Chinese physicians found that participants’ ability to understand ML decision support

tools correlated with their attitudes of usefulness of ML but not with whether they would actually use ML tools.³⁷ In studies of non-ML decision support systems, increased physician skepticism about the quality of the underlying evidence led to reduced motivation to use them.^{38,39} Finally, a qualitative study and several opinion articles from physicians outlined their desire to understand an ML’s logic before they would follow its recommendations, especially if its recommendations were to differ from their own opinion.^{24,40,41} Our study also demonstrated this trend, with no physician reporting that they would follow the ML output if they rated their understanding as “not at all.” Whilst this evidence base is small and heterogeneous, it suggests an association between physician understanding of ML outputs and intended physician behavior.

Explainability

Despite the legal requirements outlined in the Global Data Protection Regulation and the opinions described previously, many experts debate the necessity of ML explainability. Those in favor of explainability reason that even highly performant ML models are prone to error, such as confounding in their derivation datasets or incorrect application of datasets to new populations.^{24,41,42} Therefore, providing more information about the logic underlying an ML output for the physician to verify may reduce the incidence of these errors and, incidentally, may also address concerns of physician deskilling from automation complacency.^{40,42}

However, others point out that the enormous volume and complexity of pathophysiological and medical evidence involved in clinical decision making is far beyond the capacity of any physician to recall, even if they are able to review relevant information at the bedside using decision aids.²³ Furthermore, evidence used to support clinical decision making is often empirically derived from randomized clinical trials, without a pathophysiological understanding of why a treatment works.²³ This means that the current standard of care involves physicians trusting the results of trials in which they do not completely understand each component that influenced the final result. This limits their ability to ensure that patients are fully truly informed about healthcare decisions.^{23,43} Therefore, if ML models were to be incorporated into clinical practice at this standard, they may not necessarily need a comprehensive explanation of the logic underlying their output.²³

Seventy-six percent of physicians in our survey reported that the ML output was trustworthy when no model-agnostic explanation was provided. This suggests that the “empirical” evidence reported by the ML risk calculator—that the patient had a low risk (<1%) of pulmonary embolism based on a cohort of 10 000 patients who were investigated for pulmonary embolism—was often adequate to

These factors highlight the many complexities involved in the implementation of ML risk calculators into clinical practice. Although important, the purpose of our study was not to comprehensively quantify these factors, as this would have made our survey impractical. Rather, the emphasis of our study was exploring physicians' responses to contemporary model-agnostic explanations.

CONCLUSIONS

This study has found that in the healthcare setting, physician understanding, explainability, and trust in ML are related. Physicians preferred model-agnostic explanations rather than having no explanation of ML logic. However, many physicians were willing to trust a highly performant ML in the absence of an explanation of its logic. This demonstrates a discrepancy between what physicians and the law may deem to be "sufficient understanding" of ML logic when applied in healthcare settings. Further research is needed to validate these findings.

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AUTHOR CONTRIBUTIONS

WKD and NB designed the study (along with QT and RR), oversaw data collection and analysis, contributed to data interpretation, and drafted the manuscript. NH performed data analysis, contributed to data interpretation, and revised the manuscript for important intellectual content. QT designed the study (along with WKD, NB, and RR), performed data analysis, contributed to data interpretation, and revised the manuscript for important intellectual content. GS contributed to data interpretation and revised the manuscript for important intellectual content. RR is the senior investigator of the study, designed the study (along with WKD, NB and QT), contributed to data interpretation, and revised the manuscript for important intellectual content.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

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CONFLICT OF INTEREST STATEMENT

WKD and NB were contracted by Precision Driven Health to conduct the study. NH, QT, and RR were employed by Orion Health. Precision Driven Health had the right to review the publication before submission, however, the researchers had full access to the data at all times and no changes were made to the researchers' version of the manuscript.

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